COW

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June 6, 2026

1 Introduction

In this report, we present the design and evaluation of COW, an autonomous agent for the Standard track of the Supply Chain Management League (SCML), which is designed to maximize its profit through optimal resource allocation based on past trading performance and flexible price negotiations. COW is based on the strategy of "AS0," the winning agent in the Standard Track of the previous year, with improvements made to two main aspects: the "selection of major partners" and the "pricing strategy."

The dynamic acceptance-threshold adjustment mechanism implemented in AS0 is an effective approach that flexibly adapts to changes in the current negotiation day and inventory status, thereby enabling stable agreement formation. Since COW inherits this mechanism without modification, a detailed explanation is omitted in this report. This paper focuses on the main changes from AS0, namely the "selection of major partners" and the "pricing strategy," explaining them in detail with mathematical formulas.

2 Strategy

The most significant changes from AS0 are the inclusion of how favorable past transaction prices have been to the agent (price score) in the evaluation of trading partners, and the introduction of a new sequential concession logic (counter-offer strategy) linked to the negotiation status of each partner.

2.1 Selection of Major Partners

AS0 evaluated partners primarily based on the "negotiation success rate," which indicates the proportion of past negotiations that resulted in successful transactions, and allocated demand preferentially to promising partners with high success rates. In contrast, COW introduces a new "price score" that relatively evaluates how favorably a transaction was concluded in the past. It then calculates the evaluation score for each partner by combining the negotiation success rate and the price score to determine the allocation of demand. This section explains this process in two stages: the "calculation of the evaluation score" and the "demand allocation rule based on the calculated score."

2.1.1 Calculation of the Evaluation Score

Each partner i is evaluated on two axes: the "negotiation success rate" and the "price score" in past negotiations. The comprehensive evaluation score $S(i)$ takes a value between 0.0 and 1.0, where a larger value indicates that the partner is easier to negotiate with and more concessive. $S(i)$ is calculated by the following formula.

$$S(i) = \frac{W_s R_{\text{success}}(i) + W_i S_{\text{price}}(i)}{W_s + W_i} \tag{1}$$

Here, each variable is defined as follows:

- **Negotiation Success Rate ($R_{\text{success}}(i)$):** This is the ratio of successful transactions in past negotiations with partner (i), and is calculated as follows.

  Number of successful negotiations with partner i / Total number of negotiations. (In the initial state with no data, $R_{\text{success}}(i) = 0.5$).

- **Price Score ($S_{\text{price}}(i)$):** An indicator of price favorability based on the average successful transaction price P_{avg} in past transactions with partner i. We define the minimum negotiable price as P_{min}, the maximum price as P_{max}, and the range of negotiable prices as $M = P_{\text{max}} - P_{\text{min}}$.

  If the agent acts as a seller, it is advantageous to trade at a high price, so it is calculated by the following formula:

$$S_{\text{price}}(i) = \frac{P_{\text{avg}} - P_{\text{min}}}{M} \tag{2}$$

  Conversely, if the agent acts as a buyer, it is advantageous to trade at a low price, so it is calculated by the following formula:

$$S_{\text{price}}(i) = \frac{P_{\text{max}} - P_{\text{avg}}}{M} \tag{3}$$

- **Weights (W_s, W_i):** W_s represents the weight for evaluating the negotiation success rate, and W_i represents the weight for evaluating the price score. In the implementation, to slightly prioritize reliable transaction completion, we set $W_s = 0.8$ and $W_i = 0.2$.

2.1.2 Demand Allocation Rule

COW does not distribute its limited resources (demand or supply capacity) equally among all opponents. Instead, it selects promising partners who are likely to conclude transactions and generate profit as "major partners," and preferentially allocates trading quantities to them.

Let n be the total number of partners and p_{today} be the target distribution ratio of major partners to the total. The actual number of major partners k selected is determined by the following formula. Note that the specific calculation method of p_{today} directly follows the strategy of AS0, so a detailed explanation is omitted here, but it is a value that dynamically fluctuates according to the remaining time and current inventory level.

$$k = \max\left(1, \lfloor n \cdot p_{\text{today}} \rfloor\right) \tag{4}$$

Based on this k, the top k partners in descending order of the evaluation score $S(i)$ calculated in the previous section are selected as major partners. Furthermore, to prevent becoming too fixated on specific opponents (to ensure exploratory behavior), a maximum of 2 candidates are randomly selected and added from the remaining pool, forming the final negotiation partner set.

For the actual demand allocation to the selected partner set, a minimum quantity of "1" is first allocated to all of them definitely. The remaining demand is then allocated proportionally based on weights corresponding to their score rankings. This makes it possible to concentrate resources intensively on the most promising major partners while conducting efficient negotiations in parallel with multiple opponents.

2.2 Pricing Strategy

While the pricing setting in AS0 was a relatively static approach based on the median value, COW implements a rule-based logic that switches between bullishness and compromise depending on each opponent's reaction.

2.2.1 First Offer

Regardless of the opponent, in the first offer, the agent presents the most favorable price to itself, as shown below, taking an approach to explore the opponent's margin for concession to the limit.

- When acting as a seller: Maximum price P_{max}
- When acting as a buyer: Minimum price P_{min}

2.2.2 Sequential Concession in Counter Offers

In counter offers, the price is gradually conceded by a certain step width based on the number of counter offers r_{i} made to partner i. The concession width at each step is defined as 30% of the market price range M.

$$\text{concession(i)} = \min\left(M,\ M \times 0.3 \times r_{\text{i}}\right) \tag{5}$$

The negotiation price P_{trade} is determined from this concession amount and the ideal price proposed initially. When acting as a seller, the price is gradually lowered from the initial P_{max}.

$$P_{\text{trade}} = \max\left(P_{\text{min}},\ P_{\text{max}} - \text{concession(i)}\right) \tag{6}$$

When acting as a buyer, the negotiation price is gradually raised from the initial P_{min}.

$$P_{\text{trade}} = \min\left(P_{\text{max}},\ P_{\text{min}} + \text{concession(i)}\right) \tag{7}$$

With this mechanism, negotiations start from a one-sided bullish proposal initially, and a flexibility is guaranteed to quickly compromise to a realistic price for opponents with whom negotiations are prolonged.

3 Evaluation

In our experiments, we evaluated the score and stability of COW by comparing it with multiple agents in the environment of The SCML Standard 2024. Seven agents were used for comparison, including last year's winning model `AS0`, `BetterAgent`, `AdaptiveAgent`, `LearningAgent`, `KATSUDONAgent`, `PriceTrendStdAgent`, and `UltraSuperMiracleSoraFinalAgentZ`.

The number of different world configurations (n_configs) was set to 10. The experiments were conducted for three different numbers of simulation days (n_steps): 50, 100, and 200. The evaluation metric used is the average score obtained by each agent.

Table 1 shows the experimental results. The proposed method, COW, achieved the highest score among all numbers of simulation days, outperforming AS0, the winning model of last year. From these results, we conclude that the concentration of resources on promising partners by evaluating transaction records with price scores, and the introduction of a sequential concession logic for counter offers, are effective methods for improving the baseline AS0 agent.

Table 1: Experimental Results - Scores (Mean)

Agent	n_steps=50	n_steps=100	n_steps=200
COW	**0.7693**	**0.9134**	**0.9386**
AS0	0.7594	0.9002	0.9360
BetterAgent	0.6540	0.9125	0.8816
PriceTrendStdAgent	0.7454	0.8983	0.9040
AdaptiveAgent	0.6132	0.8857	0.8325
LearningAgent	0.6074	0.8639	0.7621
UltraSuperMiracleSoraFinalAgentZ	0.4644	0.9053	0.7941
KATSUDONAgent	0.4006	0.6582	0.5164

4 Conclusion

In this report, we designed a new agent, COW, which pursues further profit by improving the scoring system and introducing a step-by-step price concession, based on AS0, and explained its strategy and advantages.

COW inherits the excellent environmental adaptation functions, such as the dynamic threshold adjustment, possessed by AS0. It introduces two major improvements: the ”selection of major partners” taking into account the favorability of the transaction price, and a ”pricing strategy” that starts by proposing an ideal price and gradually widens the compromise margin depending on the number of counter offers. Experimental results in a competitive format recorded scores slightly higher than AS0 without exception in various simulation day scenarios such as 50, 100, and 200, proving that the introduction of these features contributes to a stable profit increase.